MILITARY SPECIFICATION

AIRCRAFT FIRING CIRCUIT TEST SET AN/AWM-54

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for flightline use and shall be used to perform stray energy and firing circuit tests of bomb release and missile firing circuits of Navy aircraft.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Drawing or Model Number	Nomenclature	Reference Paragraph
TS-3021/AWM-54 (P/N 178AS110)	Aircraft Firing Circuit Test Set	3.7.1
CY-6901/AWM-54 (P/N 178AS175)	Test Set Case	3.7.8
P/N 178AS200	Battery Pack	3.7.6
W-1 of AN/AWM-54 (P/N 178AS420)	Cable Assembly	3.7.9
P/N 178AS223	Adapter Retainer Assembly	3.7.10
NAVAIR 16-30AWM54-1	Technical Manual	GFM

1.3 Associated Equipment - The test set shall operate with the weapons release systems of Navy ordnance carrying aircraft.

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1.3.1 Adapter Assemblies - The adapter assemblies required for operation of the test set are listed on drawing 30003-178AS100 and the Application Data Material Readiness List (ADMRL). The adapter assemblies are not supplied with the test set and must be ordered as required.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life, for Aircraft Electronic Equipment, Format for
MIL-T-21200	Test Equipment for Use with Electronic and Electrical Equipment, General Specification for

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-756	Reliability Prediction
MIL-STD-781	Reliability Tests, Exponential Distribution

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MIL-STD-810	Environmental Test Methods
Naval Air Systems Command	
DL 178AS100	Data List for Test Set, Aircraft Firing Circuit AN/AWM-54 and all drawings listed therein
PUBLICATIONS	
AR-34	General Requirement for Failure Classification for Reliability Testing
ANA Bulletin No. 400	Airborne Electronic and Associated Equipment, Applicable Documents
MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	Maintainability Prediction

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Testing - This specification makes provisions for testing (see 4).

3.2 Parts and Materials - The parts and materials shall be in accordance with the following general and supplemental requirements.

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipment for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

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3.2.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-T-21200.

3.2.2 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-T-21200. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.3 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 178AS100 and related lists, except insofar as errors of ommission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

3.3.1 Total Weight - The total weight of the equipment, including cables, shall not exceed 36 pounds.

3.4 Availability - The availability shall be a minimum of 99 percent.

3.4.1 Reliability - Equipment reliability shall conform to the following.

3.4.1.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 100 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating Life - The equipment shall have a minimum total operating life of 1000 hours or 1,200,000 test operations with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.4.1.3 Reliability in Mean-Time-Between-Failure (MTBF) - The equipments shall demonstrate a minimum of 100 hours of mean (operating) time between failures when tested and accepted using the 1200 test operations per accumulated test hour equipment on-off cycling rate as outlined under the requirements of 4.4.3.

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3.4.2 Maintainability - Not applicable.

3.4.3 Probability of Successful Performance of Specified Test (Compatibility) - Not applicable

3.4.4 Interference Control - The test set is intended to function properly and survive without degradation or damage to interior component parts when subjected to electromagnetic environments existing on flight decks of aircraft carriers. In addition, electromagnetic radiation and susceptibility requirements of MIL-STD-461, Class 1c shall be met except as follows:

Test Code	Extended Limits	Frequency Range
RE02	60 db	14 KHz to 5 MHz

3.4.5 Maintenance Provisions and Field Testing - Not applicable.

3.4.6 Nomenclature and Nameplates - Nomenclature and nameplates shall be as defined on the drawings listed in DL 178AS100.

3.4.7 Service Conditions - The equipment shall operate satisfactorily under any environmental service conditions or combination of these conditions as specified in MIL-T-21200 for class 2 equipment, except as modified herein.

3.4.7.1 Temperature - Altitude - Testing shall be modified in accordance with the following.

3.4.7.1.1 Test Set TS-3021/AWM-54 with battery pack (P/N 178AS200) installed shall meet all requirements when tested to the limited temperatures of 0 to 55°C. Test Set TS-3021/AWM-54 with battery pack (P/N 178AS200) installed shall meet all electrical requirements for a minimum of 30 minutes when tested at temperatures of 0 to -40°C when the following conditions are met:

(a) Test Set TS-3021/AWM-54 less battery pack (P/N 178AS200) shall be stabilized at required test temperature.

(b) Battery pack (P/N 178AS200) shall be charged at 60 ma for 16 hours at standard conditions.

(c) Battery pack (P/N 178AS200) shall be placed in Test Set TS-3021/AWM-54 immediately after charging and the tests conducted after the battery has been subjected to the test temperature for a minimum of 15 minutes.

3.4.7.1.2 The battery pack shall meet the requirements of drawing 30003-178AS200.

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3.4.7.1.3 Test Set, TS-3021/AWM-54, shall be capable of unrestricted compliance with the MIL-T-21200 operational temperature - altitude requirements for class 2 equipment when powered from the external power source noted in paragraph 3.4.8 in lieu of battery pack, P/N 178AS200.

3.4.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30 ± 10°C)
Altitude	Normal ground (10,000 ft maximum)
Vibration	None
Humidity	Room ambient (up to 90 pct. relative humidity)
Input power	Fully charged battery pack (P/N 178AS200) or external power source capable of producing 200 mA of current at 11.3 to 14.0 Vdc with a maximum ac voltage of 1 mV rms.

3.5 Transportability - Not applicable.

3.6 Performance - The performance of the equipment shall be as set forth below plus any and all performance requirements stated on the drawings of DL 178AS100. Unless otherwise specified, these requirements shall apply under both standard and service conditions.

3.7 Detail Requirements

3.7.1 Aircraft Firing Circuit Test Set TS-3021/AWM-54 - The test set shall be manufactured in accordance with drawing 30003-178AS110 and all documents listed thereon. Test Set TS-3021/AWM-54 shall meet the requirements stated on drawing 30003-178AS110 and in the following subparagraphs.

3.7.1.1 Function - The test set shall operate in four functional modes; SELF TEST NO GO, SELF TEST GO, stray voltage (S/V) and firing circuit (F/C).

3.7.1.1.1 SELF TEST NO GO Mode - The test set shall provide a red NO GO lamp indication and the internally generated voltage shall be 75 to 90 mVdc when measured at the P2 end of cable W-1, P2-1 and P2-2 tied together (+), and P2-4 and P2-5 tied together (-).

3.7.1.1.2 SELF TEST GO Mode - The test set shall provide a green GO lamp indication and the internally generated voltage shall be 35 to 50 mVdc when measured at the P2 end of cable W-1, P2-1 and P2-2 tied together (+), and P2-4 and P2-5 tied together (-).

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3.7.1.1.3 Stray Voltage Test, GO Indication - The test set shall provide a green GO lamp indication when placed in the S/V mode, energized and stimulated with one of the following inputs between P2-1 and P2-4 of cable W-1:

(1) DC voltages from 0 to ± 50 mV.

(2) A single 10 μs wide rectangular pulse* from 0 to ± 6.1 V amplitude.

(3) A single 100 μs wide rectangular pulse* from 0 to ± 1.8 V amplitude.

(4) A single 1 ms wide rectangular pulse* from 0 to ± .50 V amplitude.

(5) A single 10 ms wide rectangular pulse* from 0 to ± .09 V amplitude.

(6) Sine wave voltages** from 100 Hz to 100 KHz and varying in amplitude from 0 to 55 mV rms.

(7) Simulated Blown Fuse Test. With fuse F1 removed, a dc voltage of ± 400 mV or less.

Note

*Pulse may be positive or negative. Rise and fall times shall not be greater than 1 μs.
**Total distortion of sine wave voltages shall not exceed 5 pct.

3.7.1.1.4 Stray Voltage Test, NO GO Indication - The test set shall provide a red NO GO lamp indication when placed in the S/V mode, energized and stimulated with one of the following inputs between P2-1 and P2-4 of cable W-1:

(1) DC voltages of ± 75 mV to ± 2.0 V.

(2) A single 10 μs wide rectangular pulse* of ± 10.6 to ± 150 V amplitude.

(3) A single 100 μs wide rectangular pulse* of ± 3.4 to ± 50 V amplitude.

(4) A single 1 ms wide rectangular pulse* of ± 1.1 to 15 V amplitude.

(5) A single 10 ms wide rectangular pulse* of ± .50 to ± 4.5 V amplitude.

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(6) Sine wave voltages** from 100 Hz to 10 KHz with an amplitude of 83 mV rms to 1.5 V rms.

(7) Sine wave voltages** from 10 KHz to 100 KHz with a minimum amplitude varying linearly from 83 mV rms at 10 KHz to 200 mV rms at 100 KHz.

(8) Simulated Blown Fuse Test. With fuse F1 removed a dc voltage of ± 750 mV to ± 31 V.

Note

*Pulses may be positive or negative. Rise and fall times shall not be greater than 1 μs.
**Total distortion of sine wave voltages shall not exceed 5 pct.

3.7.1.1.5 Firing Circuit Test - The test set shall check the output voltage and the output impedance of aircraft firing circuits with voltages of 24 through 31 Vdc and 19 through 26 Vac, 400 ± 40 Hz. See Figure 1 for test circuit.

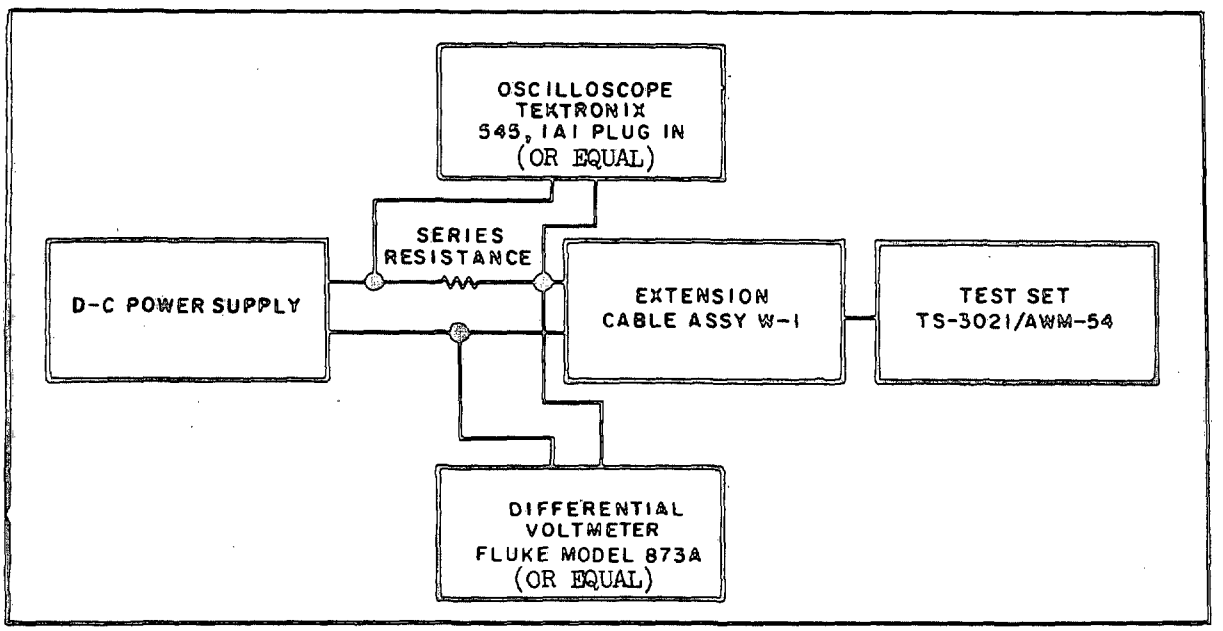


Figure 1. Test Circuit for Firing Circuit Test

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3.7.1.1.5.1 Trigger Voltage - The test set shall momentarily draw a single dc current pulse or a single train of ac current pulses as applicable which conform to the requirements of paragraph 3.7.1.5.3 and the test set shall provide a green GO lamp indication when placed in the F/C mode, energized and stimulated with 24.0 Vdc or greater, or 19.0 Vac or greater. The stimuli shall be applied at the P2 connector of cable W-1 (P2-2 (+) and P2-4 (-)). The resistance of the ac or dc source shall not exceed 5.0 Ω. The test set shall not draw current in excess of 50mA (ac or dc) and the test set shall provide a red NO GO lamp indication when placed in the F/C mode, energized and stimulated with 23.2 Vdc or less, or 16.4 Vac or less at the P2 connector of cable W-1 (P2-2 (+) and P2-4 (-)). The resistance of the ac or dc source shall not exceed 5.0 Ω. See Figure 1 for test circuit.

3.7.1.1.5.2 Source Resistance - The test set shall provide the appropriate GO or NO GO indications when placed in the F/C mode, energized and stimulated by the source voltages and output resistances as shown in Table 1 below (+ to pin 2, - to pin 4). See Figure 1 for test circuit.

Table 1. Source Voltage and Source Resistance Specifications

Lamp Indication	Source Voltage	Source Resistance*
GO	24.0 - 31.0 vdc	7.5Ω maximum
NO GO	24.0 - 31.0 vdc	8.5Ω minimum
GO	19.0 - 26.0 vac 400 ±40 Hz	7.0Ω maximum
NO GO	19.0 - 26.0 vac 400 ±40 Hz	9.0Ω minimum

*Source resistance includes the output resistance of the ac or dc power source, the series resistance of Figure 1, and all wiring and connection resistances up to the J2 connector of cable W-1.

3.7.1.1.5.3 Current Waveform - Current pulses of the F/C load circuit shall be initiated as noted in paragraph 3.7.1.1.5.1 and shall conform to the requirements of Table 2 and Figure 2. See Figure 1 for test circuit.

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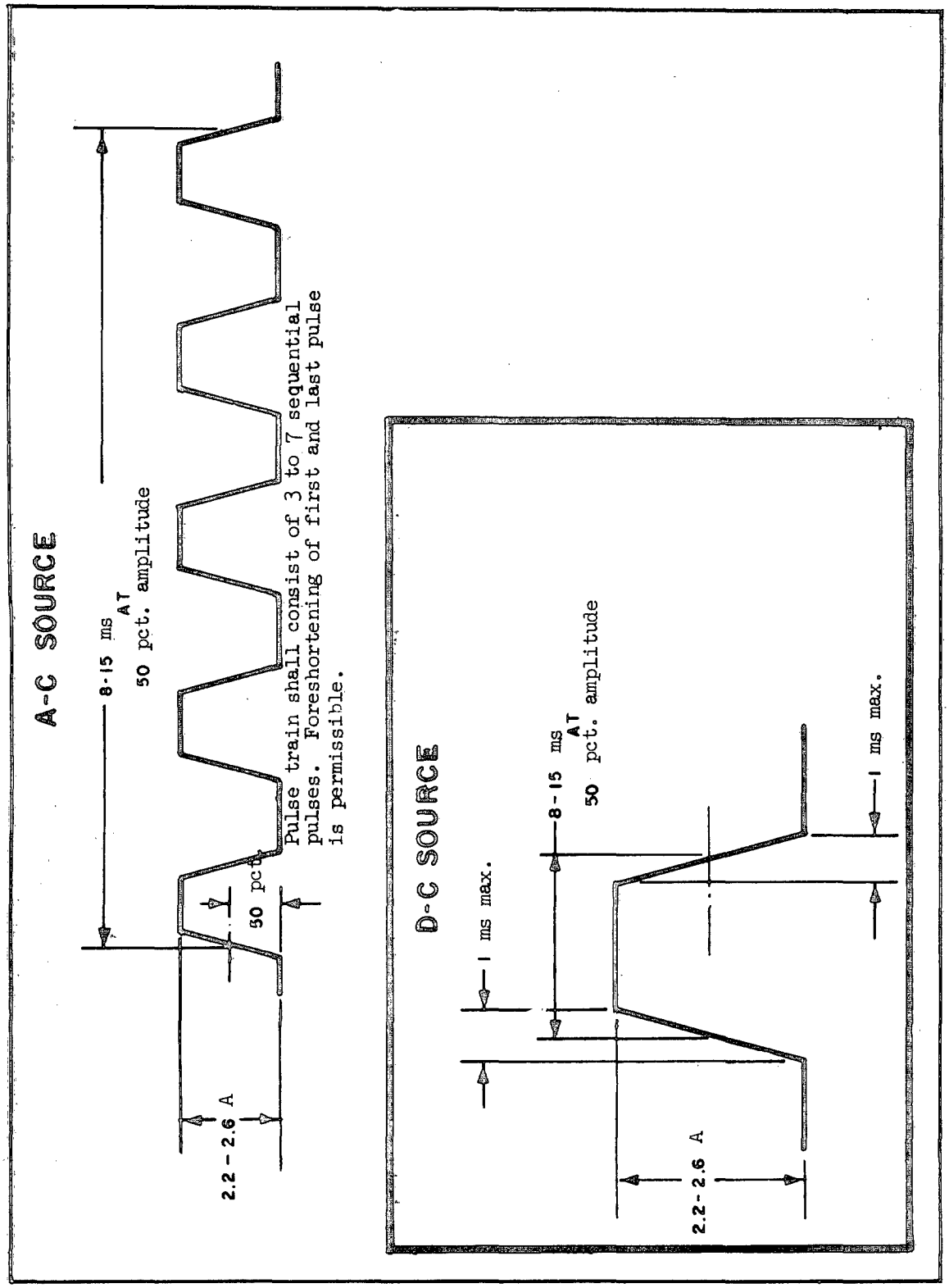


Figure 2. Current Pulse Waveforms

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Table 2. Current Pulse Specifications

Source Voltage	Pulsewidth*	Pulse Amplitude*	Source Resistance**
24.0 - 31.0 vdc	8 - 15 ms	2.2 - 2.6A	0 - 9.0 Ω
24.0 - 31.0 vdc	8 - 15 ms	2.6A maximum	9.0 Ω minimum
19.0 - 26.0 vac 400 ± 40 Hz	8 - 15 ms	2.2 - 2.6A	0 - 9.0 Ω
19.0 - 26.0 vac 400 ± 40 Hz	8 - 15 ms	2.6A maximum	9.0 Ω minimum

*See Figure 2
**Source resistance includes the output resistance of the ac or dc power source, the series resistance of Figure 1, and all wiring and connection resistances up to the J2 connector of cable W-1.

3.7.2 Primary Input Power Requirements - The power required shall be one of the following types and shall not exceed the specified amounts.

A Fully Charged Battery Pack, which meets the requirements of drawing 30003-178AS200.

An isolated dc power source capable of producing 200 mA of current at 11.3 to 14.0 Vdc, with a maximum ac voltage of 1 mV rms.

3.7.3 Warm-Up Time - The required lamp indication shall be obtained within 75 ms after the Test or Self Test switch is actuated. (See Table 3).

Table 3. Switch Positions and Require Lamp Indications

INPUT at P2 of W1	FUNCTION	PUSHBUTTON		LAMP		NOTES
		TEST	SELF-TEST	GO	NO GO	
0 to ±50 mV dc	S/V	X		X		
±75 mV to ±2.0 Vdc	S/V	X			X	
0 to 55 mV rms 100 Hz to 100 KHz	S/V	X		X		
85 mV to 1.5 v rms 100 Hz to 10 KHz	S/V	X			X	
200 mV rms to 1.50 V rms 10 KHz to 100 KHz	S/V	X			X	

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Table 3. (Continued)

INPUT at P2 of W1	FUNCTION	PUSHBUTTON		LAMP		NOTES
		TEST	SELF-TEST	GO	NO GO	
0 to ±6.1 V 10 µs pulse	S/V	X		X		
±10.6 to ±150V 10 µs pulse	S/V	X			X	
0 to ±1.8 V 100 µs pulse	S/V	X		X		
±3.4 to ±50 V 100 µs pulse	S/V	X			X	
0 to ±.50 V 1 ms pulse	S/V	X		X		
±1.1 to ±15 V 1 ms pulse	S/V	X			X	
0 to ±90 mV 10 ms pulse	S/V	X		X		
±500 mV to ±4.5 V 10 ms pulse	S/V	X			X	
0 to ±400 mV dc	S/V	X		X		Fuse F1 Removed
±750 mV to ±31 V dc	S/V	X			X	Fuse F1 Removed
0 to ±23.2 Vdc	F/C	X			X	
24.0 to 31.0 Vdc R Source 7.5 Ω or less	F/C	X		X		
24.0 to 31.0 Vdc R Source 8.5 Ω or more	F/C	X			X	
0 to 16.4 Vac	F/C	X			X	
19.0 to 26.0 V rms 400 ±40 Hz R Source 7.0 Ω or less	F/C	X		X		
19.0 to 26.0 V rms 400 ±40 Hz R Source 9.0 Ω or more	F/C	X			X	

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Table 3. (Continued)

INPUT at P2 of W1	FUNCTION	PUSHBUTTON		LAMP		NOTES
		TEST	SELF-TEST	GO	NO GO	
35 to 50 mV dc (internally generated)	GO		X	X		P2-1 & P2-2 shorted & P2-4 & P2-5 shorted at W1
75 to 90 mV dc (internally generated)	NO GO		X		X	

3.7.4 Weight - The weight of the Test Set TS-3021/AWM-54 shall not exceed 7.75 pounds.

3.7.5 Content - The Test Set TS-3021/AWM-54 contains the following major assemblies and plug-in units.

(1) Chassis Assembly P/N 178AS135

(a) Printed Wiring Board Assembly P/N 178AS112

(b) Printed Wiring Board Assembly P/N 178AS114

(2) Battery Pack P/N 178AS200

(3) Cable Assembly W-1 P/N 178AS420

3.7.6 Battery Pack P/N 178AS200 - The battery pack shall be manufactured in accordance with drawing 30003-178AS200 and shall meet the requirements stated thereon.

3.7.7 Printed Wiring Board Assemblies - The printed wiring board assemblies, P/N 178AS112 and P/N 178AS114 shall be manufactured in accordance with the documents listed on drawing 30003-178AS112 and 30003-178AS114 and shall meet the requirements stated thereon.

3.7.8 Case, Test Set CY-6901/AWM-54 - The case shall be manufactured in accordance with the documents listed on drawing 30003-178AS175 and shall meet the requirements stated thereon.

3.7.9 Cable Assembly, W1 - Cable assembly, W-1 of AN/AWM-54, connects the Test Set TS-3021/AWM-54 to the accessory adapter assemblies and shall be manufactured in accordance with drawing 30003-178AS420 and shall meet the requirements stated thereon.

3.7.10 Adapter Retainer Assembly - The adapter retainer assembly shall be manufactured in accordance with drawing 30003-178AS223 and shall meet the requirements stated thereon.

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3.8 Dimensions - The overall size of Test Set AN/AWM-54 shall be in accordance with drawing 30003-178AS174.

3.9 Finish - Finish shall be in accordance with the drawings listed on DL 178AS100.

3.10 Interchangeability - The equipment interchangeability shall be in accordance with MIL-T-21200. The drawings listed in DL 178AS100 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.11 Workmanship - Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements. Inability of the tested equipment to comply with any of the specified performance requirements shall constitute a failure:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

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4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include MIL-T-21200 environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-STD-461 to the limits of this specification.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection of an equipment and of equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - Not applicable.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test - Each Test Set TS-3021/AWM-54 shall be vibrated for a period of 10 minutes at any selected frequency within the range of 20 to 30 Hz (excluding resonant points) at a minimum amplitude of ±3 g's. The direction of vibration shall be normal to the mounting plane. Temperature and humidity shall be at room ambient. The purpose of the vibration is to detect flaws and imperfect workmanship. After vibration,

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the equipment shall again be subjected to an examination of the product and an operational test. Should a failure occur, it should be repaired and the test started over. Should repetitive failures occur, corrective action should be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. (Sampling tests shall not be conducted unless reliability assurance tests of 4.4.3 are deleted by contractual action.) Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0 (See Note)
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Note

If by contract action the initial production tests (4.3) are deleted, then a sample test shall be conducted on 1 equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests shall be conducted as required for both the Qualification Phase and Sampling Phase. Classifications of failures shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipment under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled, "Qualification Phase

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of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. The qualification phase shall be conducted in accordance with drawing 30003-178AS105.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781 under the section entitled "Sampling Phase of Production Reliability Tests." The sampling phase shall be conducted in accordance with drawing 30003-178AS106.

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during 3 months shall be one lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test are defined by drawings 30003-178AS105 and 30003-178AS106.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken by the contractor:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) If the failure is other than an isolated case:

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(a) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(b) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individua tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	One for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room ambient (30 ±10°C)
Altitude	Normal ground (0 - 10,000 ft.)
Humidity	Room ambient (up to 90 pct relative humidity)
Input. Voltage	Fully charged Battery Pack P/N 178AS200 or 11.3 to 14.0 Vdc with a maximum ac voltage of 1 mV rms, 200 mA power supply.
Vibration	None

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4.5.2 Test Periods - The test shall be run intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be put through its various phases of operation.

4.5.3 Performance Check - At approximately 24-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity every 2 weeks. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures and test set-ups used for conducting preproduction tests, acceptance test, and life test shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this

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specification and contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all "wear" or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794, Level A.

5.2 Marking - In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129. The exterior container shall be marked as follows:

> This test set contains nickel-cadmium batteries. Do not store below -40°F or above +131°F unless batteries are removed.

6. NOTES

6.1 Intended Use - The intended use of the Aircraft Firing Circuit Test Set AN/AWM-54 is to check bomb release and missile firing circuits in military aircraft for unsafe stray voltage which may cause premature firing of the bombs or missiles and to check the firing circuit output resistance and voltage to ensure that the weapons release system is in operating condition.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production

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of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than those specified in DL 178AS100, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Nonrepairable Assemblies - Not applicable.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The Drawingsof DL 178AS100 - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) ReferencedSpecifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified therein.

6.5 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.6 Type Designation - Not applicable.

6.7 Revisions - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Accessory Adapter Assemblies listed as reference on drawing 30003-178AS100.

(2) Charger Set, Battery PP-6681/AWM (P/N 178AS700).

(3) The weapons release systems of Navy ordnance carrying aircraft.

6.9 Specification Cognizance - This specification is under the cognizance of AIR-5342.

6.10 Definitions - Not applicable.

Custodians

Navy - AS

Preparing Activity

Navy - AS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated,, Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-A-81917(AS) AIRCRAFT FIRING CIRCUIT TEST SET AN /AWM-54

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Naval Air Systems Command
Washington, D.C. 20361

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